



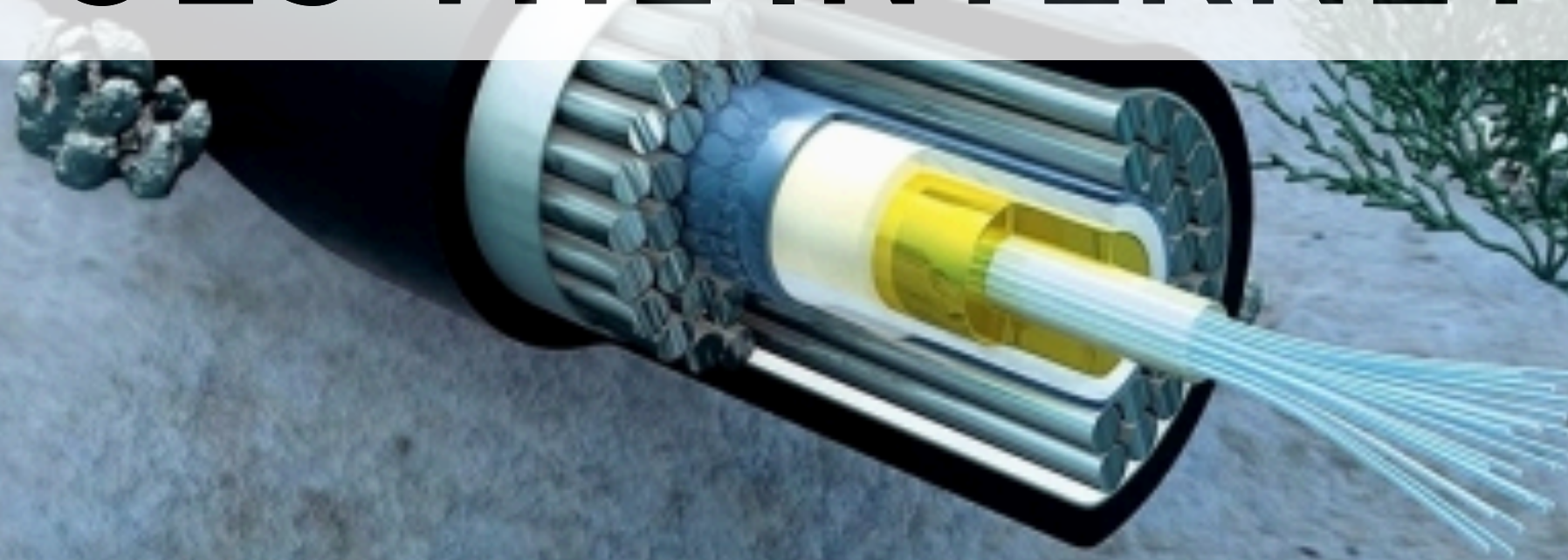
EAST TEXAS A&M
— UNIVERSITY —

CSCI 380.61E WEB PROGRAMMING AND INTERFACE DESIGN

OVERVIEW

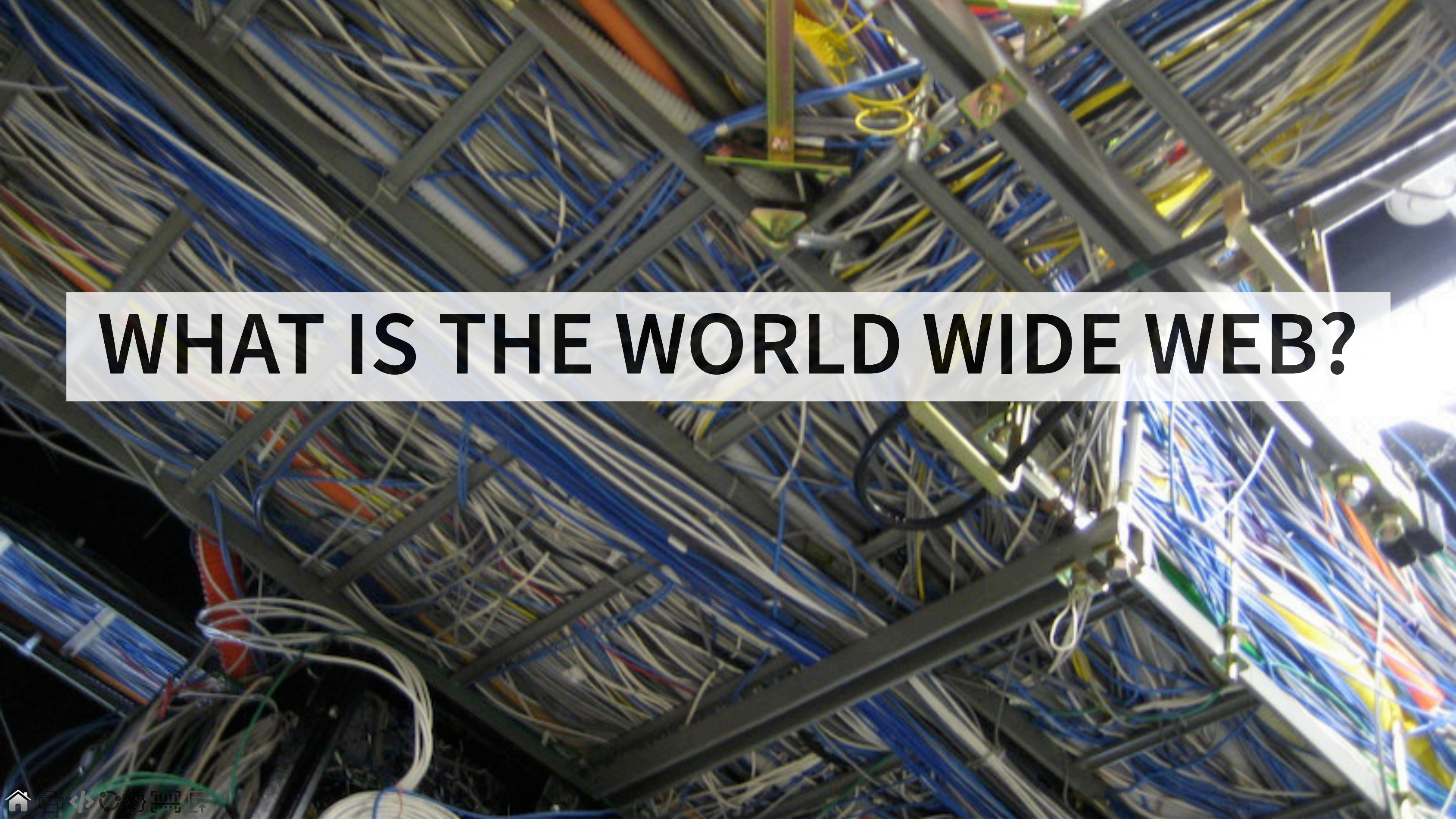
- History of the Internet
- How the Internet and the Web has Evolved
- Standards and Protocol
- Questions and Discussion

HOW DOES THE INTERNET WORK?



HOW DOES THE INTERNET WORK?

- Worldwide set of agreed common **standards** (e.g., TCP/IP)
- Internet is able to **connect** computer networks across the world (worldweb)
- **Exchange data** and information



WHAT IS THE WORLD WIDE WEB?

WHAT IS THE WORLD WIDE WEB?

- World Wide Web (WWW) is **global scale**, distributed hyper media application with **Client Server** architecture
- Resources are transferred using **Uniform Resource Identifiers (URI)** and **HyperText Transfer Protocol (HTTP)**
- WWW clients and **browsers** (agreed/supported communication standards/software systems)

GeoTraceroute to:



- #1  FR - Strasbourg (0 km)
- #2  FR - Paris (376 km)
- #3  GB - London (308 km)
- #4  US - Boston (5275 km)
- #5  US - Albany (233 km)
- #6  US - Cleveland (681 km)
- #7  US - Chicago (507 km)
- #8  US - Kansas (648 km)
- #9  US - Denver (856 km)
- #10  US - Salt Lake (629 km)
- #11  US - Seattle (1108 km)
- #12  US - Missoula (633 km)

Path via: Cogent Communications
Path / real distance: 11254 / 8013 km
Countries involved: 3
View as: [Google Maps](#) - [KML](#)

[Run another traceroute](#)

Last 6 targets checked:
209.97.192.1 (33m ago)

SHOWS TRACE ROUTE

Visual Route Packets (source-destination)

Type 'hw.ac.uk'

NATURE OF THE INTERNET

- Connect and Communicate
- Internet is an internetwork of **Wide Area Networks**
 - with gateways between them
 - based upon common use of **TCP/IP** protocols
 - supporting standard application services (DNS, e-mail, web)
- Internet has **no central operations** room or global operations manager
- Each Internet host has an IP address e.g. 137.195.13.48
- Some Internet hosts have a domain name e.g. www.macs.hw.ac.uk



LOOKUP THE IP

Online IP Checker (Domain Name System DNS)

Type 'hw.ac.uk'



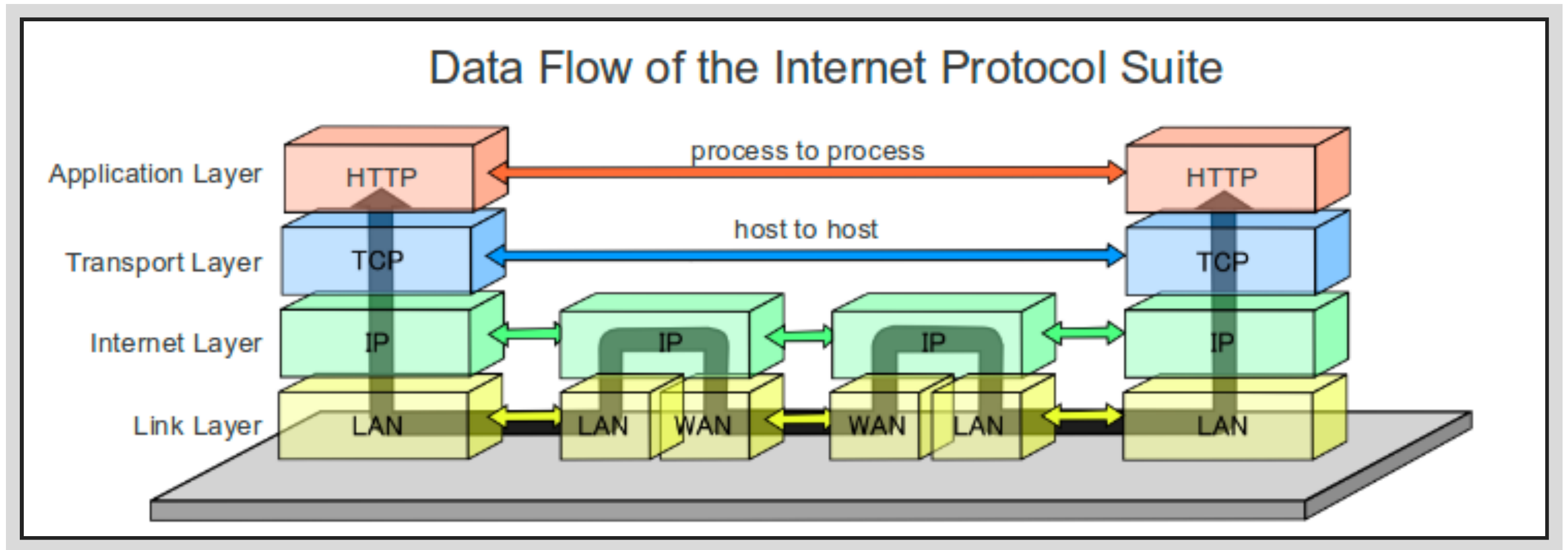
WHO REGULATES THE INTERNET?

Who regulates the internet?

- IETF, ICANN and ISOC
 - Internet Engineering Task Force (IETF)
 - Internet Corporation for Assigned Names and Numbers or ICANN
 - Internet Society (ISOC)

HOW ARE THE TCP/IP NETWORKS STRUCTURED?

TCP/IP NETWORKS CAN BE VIEWED AS 4 LAYER STRUCTURE



LAYER FUNCTIONALITY

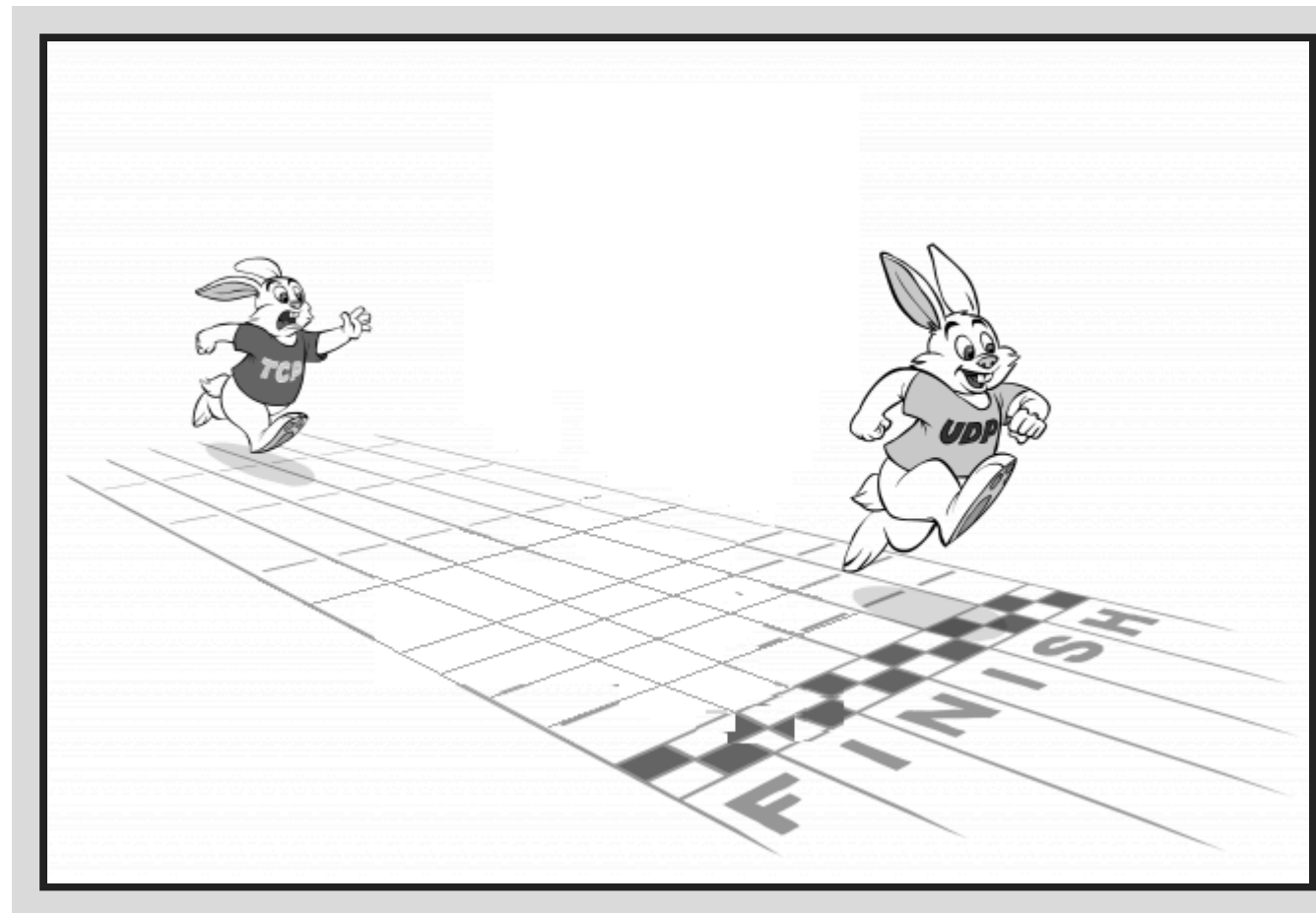
- Application interfaces directly with user applications or **users**
- Transport end to end (un)reliable delivery of **TCP** or UDP packets
- **Network** IP datagram delivery, addressing, routing
- Link delivers frames, handles errors, drives **physical transfers**

EXAMPLES OF PROTOCOLS (INTERNET LAYERS)

- Application SMTP, IMAP, **HTTP**, DNS, RTP, SNMP, TFTP
 - Transport **TCP**, UDP, SCTP
 - Network **IP**, ICMP, IPsec, IGMP
 - Link Ethernet, 802.11, DSL, ARP, L2TP, ISDN, GPRS, PPP
-
- TCP carries reliable services - email (SMTP, IMAP), **web (HTTP)**
 - UDP carries best efforts services - DNS, media streams (RTP)

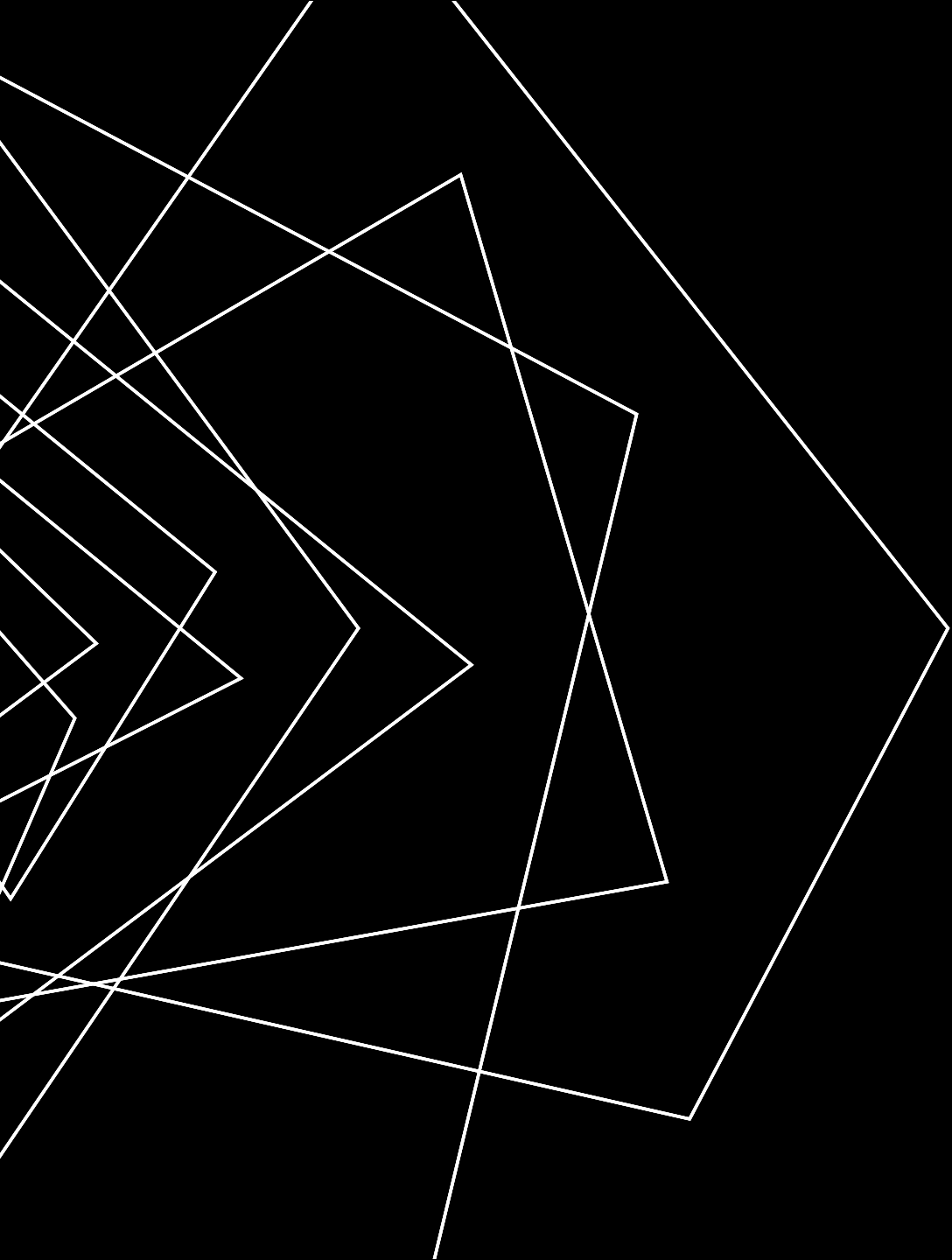
TCP OR UDP?

- What are the differences?
- When would you use TCP or UDP?



SUMMARY

- Understand Core Concepts (Web & Internet)
- Structure and Terminology
- High and Low Level View



THANK YOU

Yau Ka Cheung
Yauka.Cheung@etamu.edu
ACB1 Room 335